

Module 10: Notes

Lesson : Grid in CSS

Introduction to Grid

- **Layout System:** CSS Grid is a 2-dimensional layout system for arranging elements in rows and columns.
- **Activation:** Use `display: grid;` on a container to enable grid layout.
- **Structure:** The grid consists of a grid container (parent) and grid items (children).
- **Grid Lines:** Define column and row boundaries within the grid.
- **Grid Properties:** Includes `grid-template-columns`, `grid-template-rows`, `grid-column`, and `grid-row` for defining layout specifics.
- **Advantages:** Offers flexibility, simplicity in creating layouts, and precise control over element positioning.

Grid container properties

- **Display:** Activates grid layout on an element using `display: grid;`.
- **grid-template-columns:** Defines the number and sizes of columns in the grid layout.
- **grid-template-rows:** Specifies the number and sizes of rows in the grid layout.
- **grid-auto-rows:** Sets the size of rows that are not explicitly defined in `grid-template-rows`.
- **grid-auto-columns:** Sets the size of columns that are not explicitly defined in `grid-template-columns`.
- **grid-template-areas:** Defines named grid areas where grid items can be placed.
- **grid-column-gap:** Specifies the size of gaps between columns in the grid layout.
- **grid-row-gap:** Specifies the size of gaps between rows in the grid layout.
- **justify-content:** Aligns grid items along the main axis (horizontal) of the grid container.
- **align-content:** Aligns grid lines along the cross axis (vertical) of the grid container when there is extra space.
- **justify-items:** Aligns grid items along the inline (horizontal) axis within their grid area.
- **align-items:** Aligns grid items along the block (vertical) axis within their grid area.

Grid item properties

- **grid-column-start** and **grid-column-end**: Specifies the starting and ending positions of a grid item along the columns.
- **grid-column**: Shorthand for setting grid-column-start, grid-column-end, or both.
- **grid-row-start** and **grid-row-end**: Specifies the starting and ending positions of a grid item along the rows.
- **grid-row**: Shorthand for setting grid-row-start, grid-row-end, or both.
- **grid-area**: Defines a grid item's size and location within the grid by referencing named grid areas.
- **justify-self**: Aligns a grid item along the inline (horizontal) axis within its grid area.
- **align-self**: Aligns a grid item along the block (vertical) axis within its grid area.
- **order**: Controls the order in which grid items appear in the grid layout. Items with lower order values appear first.